



STREAKIT — MVP FEATURE SET

1. 🏠 Student Dashboard — MUST HAVE

The student's home base.

Shows:

- Current learning streak
- Experiments completed
- Overall progress
- Skills practiced
- Recent experiment/activity
- Recommended next experiment
- Daily microbiology challenge

Purpose:

Answer immediately:

Where am I? What have I done? What should I do next?

2. 🧪 Experiment Library — MUST HAVE

A dedicated microbiology experiment catalogue.

Each experiment card should show:

- Experiment name
- Short description
- Difficulty
- Estimated time
- Skill/concept
- Completion status
- Best score / mastery status

MVP experiments:

1. **Gram Staining**
2. **Streak Plate**
3. **Antimicrobial Susceptibility / Kirby-Bauer**
4. **Biochemical Testing**
5. **Microbial Identification**

I'd keep the library to **5 strong experiments** rather than pretending we need 50.

The platform's value should come from **depth of interaction**, not experiment count.

3. **Experiment Brief — MUST HAVE**

Before starting an experiment:

Student sees:

- Objective
- Core concept
- Equipment/materials
- Safety
- What they will do
- What they should learn

Then:

START EXPERIMENT

This is the bridge between learning content and simulation.

4. **Interactive Virtual Laboratory — CORE MUST HAVE**

This is the heart of StreakIt.

The student should actually **perform** the experiment.

Not watch it.

Not click through slides.

Core interactions:

- Select equipment/materials
- Select reagents/samples
- Pick correct actions
- Apply/use materials
- Follow procedure
- Make decisions
- Submit/finish experiment
- Observe simulated outcome

The exact interactions vary by experiment.

5. 🧠 Decision Points — CORE DIFFERENTIATOR

This is **non-negotiable**.

Don't make:

Click → Next → Click → Next → Done.

Instead:

What should you do now?

For example, Gram Staining:

Select reagent → apply → timing → rinse → decide next step → decolorize → observe → interpret

The student has to **think**, not just follow instructions.

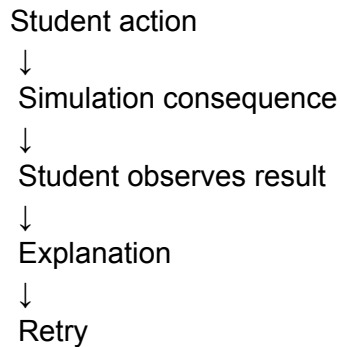
6. ⚠️ Mistake + Consequence System — CORE DIFFERENTIATOR

This might be the most important MVP feature after the simulation itself.

If the student makes a mistake:

Don't just say "Wrong."

Instead:



Example:

Over-decolorization

- cells lose expected purple staining
- student sees abnormal result
- identifies possible cause
- receives explanation
- retries

This is exactly where StreakIt starts becoming **learning-by-doing**.

7. StreakIt AI Lab Mentor — MUST HAVE

Dedicated AI assistant available inside the platform.

Not just an instruction icon.

It can:

- Explain concepts
- Explain why a step matters
- Explain equipment/reagents
- Explain mistakes
- Troubleshoot results
- Answer student questions
- Give contextual hints
- Explain observations

Critical constraint:

The AI should guide, not perform the experiment.

Student remains responsible for decisions.

8. Result + Interpretation — MUST HAVE

The experiment shouldn't end at:

Experiment Complete 🎉

The student must interpret what they produced.

For example:

Gram Stain

What is the Gram reaction?

Why did the cells produce this result?

Streak Plate

Is this plate successfully isolated?

What does the colony distribution tell you?

This moves StreakIt from **procedure simulation** → **scientific reasoning**.

9. Assessment + Scoring — MUST HAVE

At the end:

Student receives:

- Overall score
- Procedural accuracy
- Decision accuracy
- Interpretation accuracy
- Mistakes made
- Concepts needing improvement

Potentially:

Score: 84%

Strong: Reagent sequence

Needs practice: Decolorization

Interpretation: 90%

This gives the platform measurable learning outcomes.

10. Retry / Practice — MUST HAVE

The student should be able to immediately:

Retry Experiment

Especially after mistakes.

But ideally the second attempt isn't just an identical replay.

The system can preserve:

- Previous mistakes
- Previous score
- Improvement

So:

Attempt 1 → 68%

Attempt 2 → 84%

Mastered → 92%

That's where your **practice** → **mastery** concept starts becoming real.

11. Skill-Based Progress — MUST HAVE

Not just:

“You completed 4 experiments.”

Track actual skills.

For example:

Microbiology Skills

- Staining — 80%
- Aseptic technique — 65%
- Culture techniques — 72%
- Biochemical testing — 55%
- Interpretation — 88%
- Antibiotic sensitivity — 70%

This is much more meaningful than simple completion percentages.

12. Streak System — MUST HAVE

This is literally in the name.

Track:

- Daily learning streak
- Experiments/practice contributing to streak
- Current streak
- Best streak

But keep it **lightweight** for MVP.

No giant achievement ecosystem yet.

The goal is:

Come back. Practice. Improve.

13. Daily Microbiology Challenge — MUST HAVE

A small activity that gives students a reason to return even when they don't have time for a full experiment.

Examples:

Identify the Gram reaction

Choose the correct reagent

Diagnose a failed streak plate

Interpret a biochemical result

Flow:

Challenge → Answer → Explanation → Streak progress

This supports the habit loop without requiring a 20-minute lab session.

14. Learning History — MUST HAVE

The system should remember:

- Experiments attempted
- Scores
- Mistakes
- Attempts
- Skills
- Improvement
- Mastery

This feeds the dashboard and personalization.

15. Recommendation System — MVP-LITE

After an experiment:

Recommended next: Streak Plate

or:

Practice Decolorization

or:

Review: Differential Staining

It doesn't need sophisticated machine learning yet.

Rule-based recommendations are perfectly fine for MVP.



THE TWO MODES

I would also lock **two experiment modes** into the MVP.

LEARN MODE

AI Mentor available.

Hints available.

Guidance available.

Mistake explanations available.

Designed for **learning**.

TEST MODE

Minimal/no guidance.

Student performs independently.

Mistakes still affect the result.

Final score recorded.

Designed for **assessment**.

This distinction is VERY valuable.

NOT MVP

Now the brutal part.

These stay **out** for now:

Massive experiment library

PraxiLabs can win that game.

Don't copy it.

Full e-book/resource library

Useful later, not core MVP.

Instructor dashboard

Later.

Institutional administration

Later.

✗ LMS integration

Definitely later.

✗ Multiplayer labs

No.

✗ Social features

No.

✗ Leaderboards

Not necessary.

✗ Complex achievements/badges

No.

✗ Advanced adaptive learning engine

Later.

✗ Full molecular biology/genetics

Not MVP.

✗ VR/AR

Absolutely not necessary.